



affordable, and mainly recyclable materials. The researchers then used an aluminium anode to convert the gathered energy into current, which they then put into a board with an Arm Cortex M0 + CPU.

Copper 'headphones' improve atomic radio receiver

The National Institute of Standards and Technology (NIST) has improved the sensitivity of their atomic radio receiver by putting the small glass cylinder containing cesium atoms inside what appear to be special copper 'headphones.' The structure, which consists of a square overhead loop connecting two square panels, boosts the incoming radio signal (or electric field). This signal is applied to the gaseous atoms in the flask (also known as a vapour cell) between the panels. The structure is actually a split-ring resonator that functions like a metamaterial, due to which the radio receiver can now identify much weaker signals than before.



Copper headphones boost the sensitivity of NIST's atomic radio receiver, which is composed of a gas of cesium atoms prepared in a special state inside the glass container. When an antenna located above the setup sends down a radio signal, the headphones boost the strength of the received signal a hundredfold (Credit: NIST)

A seaglider that tracks ocean acidification

The University of Alaska Fairbanks and its industry partners are the first in the United States to use an unmanned underwater vehicle to monitor carbon dioxide, the primary cause of ocean acidification. The researchers designed a



Scientists and their industry partners aboard the research vessel Nanuq recover the carbon dioxide-measuring Seaglider after a test dive in the Gulf of Alaska (Credit: Heather McFarland, International Arctic Research Center)

strategy to close the data gaps that exist in the marine environment. They found a means to link a carbon dioxide sensor with a Seaglider that can dive to 1000 metres and carry out weeklong expeditions to distant sections of the ocean in any season, working with worldwide business partners. The most exciting thing is that a power-hungry sensor like the CO₂ sensor was efficiently used on a small glider that was not meant to do stuff like this!

Researchers develop artificial skin containing heat sensors

Researchers at TU/e have devised a method for producing flexible, thin, and precise sensor electronics that can help robots detect humans. Many industrial operations require close human-robot interactions. This means that the robots need accurate proximity sensors to check if someone gets



Robot arms could become safer in industrial settings by applying an artificial skin containing proximity heat sensors to detect humans in all directions (Credit: Marco Fattori)

too close to the robot. To aid this, the researchers created a large flexible sheet filled with heat sensors that can be used to cover a structure, such as a robotic arm. The flexible sensor acts like a sort of 'artificial skin' for the robot.

The heat sensors

allow the robot to detect the presence of a moveable heat source, such as a person, from a distance of up to 0.4 metres.

Soon 3D FETs will be used as memory devices

Current flash memory technologies require relatively large currents to read or write data. However, mobile cloud-enabled devices and IoT devices require memory that is energy-efficient and small in size. The University of Tokyo's Institute of Industrial Science used a ferroelectric gate insulator and an atomic-layer-deposited oxide semiconductor channel to construct 3D vertically-formed FETs for high-density data storage devices. Furthermore, they discovered that by employing antiferroelectric materials instead of ferroelectric, they were able to erase data with only a small net charge, resulting in more efficient write operations. This research could lead to new data storage memory that is even smaller and more environmentally friendly.